CONGRUENCE OF SEGMENTS

How to read the figures. —— given - - - auxiliary (added in the proof)
—— what is being proved or asked

4–1 INTRODUCTION

In this chapter we shall be concerned with comparing the "sizes" of segments. All of us have learned to understand the terms "distance" and "length" as used in everyday life. For example, we may say that it is 12 miles to the nearest town, or that a fence is 153 feet long. In each case we give a number to indicate a distance or length. To give a number is a most convenient means of describing the distance between two points or, in other words, of telling how far apart they are.

But even before men learned to count, they must have wished at times to tell how far apart points were, or to decide if two objects had the same length. For example, a hunter visiting a friend in a neighboring village might have said to his friend, "My son is as tall as this mark on my spear." Then the men could see, by comparing with the mark, whether the host's son was as tall as the visitor's son. In this example the marked spear served as a kind of "yardstick" for comparing the height of the hunter's son with the height of others. Even in very primitive societies men must have had a means for comparing the sizes of objects.

[Illustration: a boy standing on hatched ground beside a marked spear planted upright, its point above his head.]

FIGURE 4–1

Problem. Describe other examples where objects are compared in size without the use of measurement, that is, a number. Some of your examples may use a marked stick (or "yardstick") and some not.

We are so used to speaking of distance or of length in terms of numbers that it is hard for us to find other words to describe how far apart two points are. Although we usually associate numbers with the words "distance" and "length," what we study in this chapter is so simple that it would be understandable to a person who knew nothing of numbers at all. We will be talking about the "size" of a segment and whether two segments are "the same size." As we have seen with the example of the hunter, a practical method for comparing segments is to use a marked stick and to carry it from one place to another. We shall see in the Section 4–2 how in geometry we can avoid carrying such sticks.

Exercise Group 4–1

1. Your foot is a natural "yardstick" you carry with you. Describe several examples of using your foot for comparing lengths.
2. Think of several ways to use your hands as "yardsticks."

4–2 CONGRUENCE

In daily life we are often interested in size, and we have ways of testing whether or not two things are the same size. But our geometry so far has in it *no notion of size at all*. Because our geometry deals with imaginary concepts (points and lines), to talk about size it is necessary to assume (postulate) properties that we desire. We do not wish to talk about "carrying a segment around in the plane," because this involves the idea of movement, which is not a part of geometry proper. Consequently *we will just suppose that by some means (it does not matter how) we can tell whether or not two segments are the same size.* In other words, we will assume that we can pick out all the segments of the same size.

We have been using the words "the same size." We now use another word which is to have exactly the same intuitive significance. The word is *congruent*, and whenever you see it you can, if you like, say to yourself, "the same size." Congruence refers to the *relation* of being the same size. The word comes from a Latin word meaning "agree" or "conform."

Euclid used the word "equal" instead of congruent. However, *in this book we will use the word "equal" (and the sign "=") to mean "identically the same."* Thus, if we write $A = B$ we mean that A and B are one and the same point. Or, if x and y are numbers, $x = y$ means that they are one and the same number. And in dealing with segments, if we write $AB = CD$, we mean that $A = C$ and $B = D$ or $A = D$ and $B = C$, that is, AB and CD are one and the same segment.

4–3 POSTULATE GROUP III. LINEAR CONGRUENCE

We assume that there is a relation between segments called *congruence*. This relation is to have the following properties.

III–1 *The existence postulate for segments.* Given points A and B on a line l and a point A' on a line l' (Fig. 4–2), then on each side of A' on l' there is exactly one point B' such that AB is congruent to $A'B'$. To indicate that AB is congruent to $A'B'$, we write "$AB \cong A'B'$."

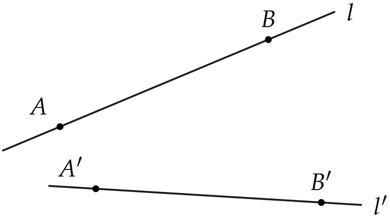

FIGURE 4–2

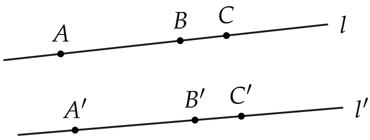

FIGURE 4–3

In congruence the order of the points does not matter, that is, if $AB \cong A'B'$, then also $AB \cong B'A'$, $BA \cong A'B'$, and $BA \cong B'A'$.

III–2 *The three-part segment postulate.* Congruence also satisfies the following:

(a) $AB \cong AB$. (A segment is congruent to itself; that is, congruence of segments is reflexive.)

(b) If $AB \cong A'B'$, then $A'B' \cong AB$. (Congruence of segments is symmetric.)

(c) If $AB \cong A'B'$ and $A'B' \cong A''B''$, then $AB \cong A''B''$. (Congruence of segments is transitive.)

Problem. Prove that if $AB \cong CD$ and $EF \cong CD$, then $AB \cong EF$. [*Hint:* Use parts (b) and (c) of III–2.]

III–3 *The addition and subtraction of segments postulate.* Suppose that B is between A and C on a line l, and that B' is between A' and C' on a line l' (Fig. 4–3).

(a) If $AB \cong A'B'$ and $BC \cong B'C'$, then $AC \cong A'C'$.

(b) If $AC \cong A'C'$ and $BC \cong B'C'$, then $AB \cong A'B'$.

These postulates are somewhat like our earlier ones. They seem so obvious that you might feel they do not need to be stated. As before, however, you must remember that points and lines are inventions of the mind, and so we must state clearly what properties they are to have. Euclid did not state them as clearly as we have. The postulates given in this book are easier to use than those of Euclid because they tell you exactly what you can do. They are similar to ones given by David Hilbert (see Section 1–2).

In less exact language, the postulates say the following things. The first says that on the line l' there are two points B' and B'', one on each side of A', such that AB is the same size as both $A'B'$ and $A'B''$. The second one says three things: that a segment is the same size as itself; that if AB is the same size as $A'B'$, then $A'B'$ is the same size as AB; and that if two segments are the same size as a third segment, then the two segments are the same size. The third postulate tells how to combine segments by what we call *addition and subtraction of segments*.

Exercise Group 4–2

1. What does it mean to say $RS = TU$?

2. If $RS = TU$, is $RS \cong TU$? Justify your answer.

3. Is it true for every pair of segments RS and TU that if $RS \cong TU$ then $RS = TU$?

4. Two segments AB and CD are not congruent to each

other. (We write this as $AB \ncong CD$.) Prove that AB and CD are not equal to each other ($AB \neq CD$).

5. If ABC, what postulates must you use to prove that $AB \ncong AC$?

6. Is the following statement true or false? If AB is any segment and A' is any point on a line l', then there is one and only one point B' on l' such that $AB \cong A'B'$.

7. In Fig. 4–4 the following congruences hold: $AB \cong BC$, $CD \cong CE$, $CE \cong ED$, $EF \cong EB$. List all other congruences that you recognize.

8. The sign $=$ was defined in this section. State the definition in your own words.

9. State the three postulates of linear congruence in your own words, using the phrase "the same size" instead of "congruent." Then go back and read again the paragraph preceding this group of exercises.

10. Show (using postulates) that if A, B, C, D are points, in this order, on a line l, and if A', B', C', D' are points, in this order, on a line l', and if $AB \cong A'B'$, $BC \cong B'C'$, $CD \cong C'D'$, then $AD \cong A'D'$. [*Hint:* Draw a figure. You will have to use part (a) of Postulate III–3 twice.]

11. Assuming that the points have the same order as in Exercise 10, prove that $(AD \cong A'D' \text{ and } CD \cong C'D' \text{ and } BC \cong B'C') \to AB \cong A'B'$.

12. Suppose E, F, G and E', F', G' are *any* points on lines l and l', respectively, such that $EF \cong E'F'$ and $FG \cong F'G'$. Show, by drawing a figure, that it does not need to be true that $EG \cong E'G'$. What do we need to know about the position of the points? Does III–3 take care of this?

*13. Part (b) of Postulate III–3 can be *proved* from the other postulates. We have included this as a postulate to make things easier. Try to prove (b). You will have to use (a) and Postulate III–1.

*14. Try to reword III–3 in terms of length.

15. Each part of Fig. 4–5 indicates certain congruence relations. The marks show which segments are congruent. What other relations can be deduced? Point out how the postulates are used.

*16. Show how to relabel the points in parts (b) and (c) of Fig. 4–5 as A, B, C and A', B', C' so that the application of Postulate III–3 is clearly shown.

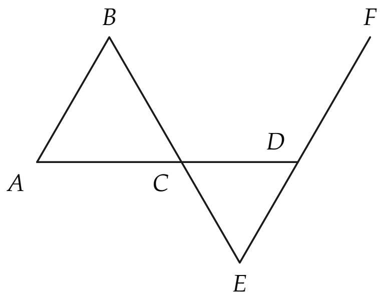

FIGURE 4–4

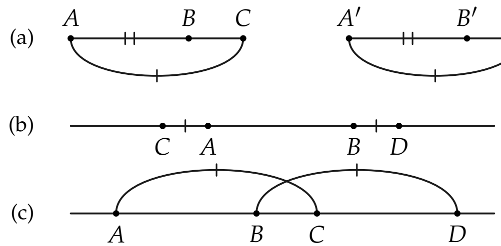

FIGURE 4–5

4–4 COMPARING SEGMENTS

The postulates in Group III tell us about segments that are the same size or congruent to each other. We can now *define*, in terms of congruence and betweenness, the ideas of larger and smaller segments. The definition will simply describe what the hunter did when he compared the boy with the mark on the spear.

We need a symbol to stand for "is greater than" and another symbol to stand for "is less than." These symbols are: $>$ and $<$. Whenever you see $>$, read "is greater than," and whenever you see $<$, read "is less than."

Definition 4–1. Let AB be any segment and $A'B'$ any other segment on line l' (Fig. 4–6). On the same side of A' as B', let B'' be the point such that $AB \cong A'B''$. If $A'B'B''$ (that is, if B' is between A' and B''), we say that $AB > A'B'$. If $A'B''B'$ (that is, B'' is between A' and B'), we say that $AB < A'B'$.

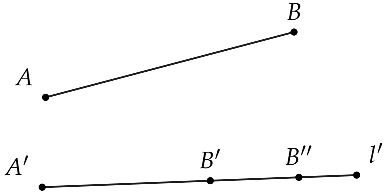

FIGURE 4–6

Remark. Because of III–1 there is only one point B'' on the same side of A' as the point B' and such that $AB \cong A'B''$. The three points A', B', and B'' are therefore in some definite order (unless $B' = B''$, in which case $AB \cong A'B'$). We cannot have A' between B' and B'' because B' and B'' are on the same side of A'. Therefore if $AB \ncong A'B'$, we must have either $A'B'B''$ or $A'B''B'$. In other words, either $AB > A'B'$ or $AB < A'B'$. This remark proves the following theorem.

Theorem 4–1. *If AB and $A'B'$ are any two segments, then either $AB > A'B'$, or $AB \cong A'B'$, or $AB < A'B'$, and exactly one of these is true.*

Exercise Group 4–3

1. Does Fig. 4–7 indicate that $AB < CD$ or that $CD > AB$? Read Definition 4–1 again.
2. What relationships are indicated by Fig. 4–8? Use Definition 4–1.
3. What relationships are indicated by Fig. 4–9? Use Definition 4–1.
4. If we know that $AB \cong CD$ and ABX, what can we conclude? Use Definition 4–1.
5. It is known that $RS \cong UT$ and RVS. What conclusion can be drawn? Use Definition 4–1.
6. Does Definition 4–1 say that if $AB > CD$ then $CD < AB$? Does looking at a figure prove that this is so?

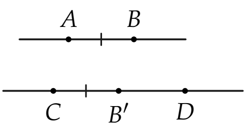

FIGURE 4–7

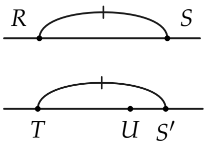

FIGURE 4–8

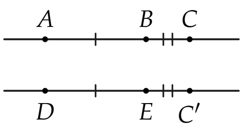

FIGURE 4–9

The exercises above are meant to prepare you for the main theorem of this section, namely: if one segment is greater than a second, then the second is less than the first. Of course, we know that if we draw two pencil marks so that the first is longer than the second, then the second is shorter than the first. But we have not yet defined *length of a segment*. As a matter of fact we shall need to introduce a new postulate into our geometry before length can be defined. Since our definitions of "greater than" and "less than" have been formulated in terms of betweenness, the truth of any statement about these relations ("greater than" and "less than") must be established by using properties of betweenness.

†**Theorem 4–2.** *If $AB > A'B'$ then $A'B' < AB$, and conversely, if $A'B' < AB$ then $AB > A'B'$.*

Proof. Figure 4–10 illustrates what must be proved in order to establish the first part of the theorem.

(1) $AB > A'B'$ means $AB \cong A'C'$ and $A'B'C'$.

(2) $A'B' < AB$ means $A'B' \cong AC$ and ACB.

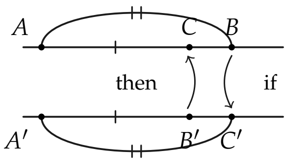

FIGURE 4–10

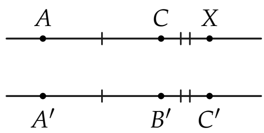

FIGURE 4–11

To show that $AB > A'B' \rightarrow A'B' < AB$, we must prove that if C' falls "beyond" B', then C falls between A and B.

From (1) above we have $AB \cong A'C'$ and $A'B'C'$. We know also that on a given side of A there is a point C such that $AC \cong A'B'$. To establish that we do indeed have ACB, consider the point X in Fig. 4–11 such that ACX and $CX \cong B'C'$. Now, by addition of segments, $AX \cong A'C'$. But $AB \cong A'C'$, so $X = B$. (Why?) And since ACX, we have ACB; and from (2) above we have $A'B' < AB$.

† Although this theorem is intuitively obvious, its proof is quite difficult. If the teacher wishes, it may be treated as a postulate.

The proof of the converse is similar to the above proof and is left as an exercise.

From now on we shall freely interchange the "greater than" and "less than" terminology since the statements $AB > CD$ and $CD < AB$ are equivalent to each other.

Theorem 4–3. *If $AC \cong A'C'$ and $A'B'C'$ and B is the point on C's side of A such that $AB \cong A'B'$, then ABC (Fig. 4–12).*

Proof. Since $AC > A'B'$, $A'B' < AC$, and so by Definition 4–1 we have B between A and C.

This theorem is proved at this time because we shall need it in the next chapter when we discuss the length of a segment. We can then show that congruent segments have the same length. It is also a very convenient tool to use in many subsequent proofs.

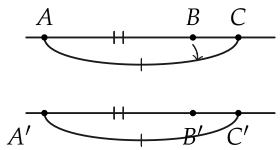

FIGURE 4–12

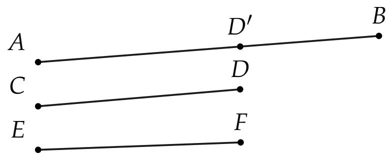

FIGURE 4–13

Definition 4–2. If we wish to say "$AB \cong A'B'$ or $AB > A'B'$," we write "$AB \geqq A'B'$" and read the symbol $\geqq$ as "is greater than or congruent to." Similarly, if we wish to say "$AB \cong A'B'$ or $AB < A'B'$," we write "$AB \leqq A'B'$" and read the symbol $\leqq$ as "is less than or congruent to."

Theorem 4–4. *If $AB > CD$ and $CD \cong EF$, then $AB > EF$.*

Proof. Since $CD < AB$ (why?) there is a point D' (Fig. 4–13) on AB such that $CD \cong AD'$. But then, by III–2, $EF \cong AD'$. Hence, by Definition 4–1, $EF < AB$ and $AB > EF$.

Exercise Group 4–4

1. Using the definition of $>$, explain what must be done in order to prove that $AB > CD$. Draw a figure to illustrate this.

2. What must one show in order to prove $RS < TU$? Draw a figure to illustrate this.

3. What basic undefined relations are used to define the phrases "is greater than" and "is less than"?

4. Is it correct to say that $AB \geqq CD \rightarrow AB \ncong CD$? Is the converse of this implication true?

5. Is it correct to say that $AB > CD \rightarrow AB \ncong CD$? State both the converse and contrapositive of this statement. Which one of these statements is true?

6. What conclusion can you draw if you know that A, B, C, and D are points on line l and that $AB > CD$ and $A = C$?

7. If AB is neither equal to nor greater than CD, can you conclude $AB < CD$?

8. If AB is neither congruent to nor less than CD, can you conclude $AB > CD$?

*9. Prove that if $AB < EF$ and $EF \cong CD$, then $AB < CD$. State the converse and contrapositive of this theorem.

*10. Prove that if $AB > CD$ and $CD > EF$, then $AB > EF$.

*11. Prove that if $AB < CD$ and $CD < EF$, then $AB < EF$.

*12. Prove that if B is between A and C, and B' is between A' and C', and $AB \cong A'B'$, and $BC > B'C'$, then $AC > A'C'$.

13. If $AB > CD$ and $CD < EF$, can you prove that $AB > EF$?

*14. Prove that if ABC and $A'C'B'$ and $AC \cong A'C'$, then $AB < A'B'$.

REVIEW OF CHAPTER 4

Several important ideas have been developed in this chapter. You should realize that *congruence* is really an undefined relation. Of course, you are so accustomed to comparing the sizes of things by comparing numbers that it is probably quite difficult to see clearly that numbers are not actually needed. We know that a $6'8''$ basketball center is taller than a $6'2''$ center. We know that a 240-lb football tackle is larger than a 180-lb tackle. But such comparisons as these can be made *without using numbers at all*. Surely men were able to recognize differences in sizes long before they had a number system to measure them. Historically, the Greeks developed their geometry without the use of number concepts.

In the next chapter, when we define what is meant by the length of a segment, it will naturally turn out that congruent segments have the same length. That is, if $AB \cong CD$, and x and y are numbers which are the lengths of these segments, then $x = y$.

Again, the postulates of this chapter are so obvious to our intuition that you might wonder why we bother to make them. All we can say is that postulates should be obvious. The amazing thing about mathematics is that out of such simple assumptions as we are making in our study of geometry one can construct theorems of marvelous complexity which provide the foundation for all work in science.

The proofs in this chapter are quite difficult. There is no need to memorize the results of the theorems since they merely show that our geometry agrees with the figures that we draw.

In working with Definition 4–1, you should have learned an important mathematical lesson, namely, that definitions must be clearly understood. It would be very easy to read Definition 4–1 carelessly and to assume that if $AB > CD$ then $CD < AB$. This is not stated in the definition, but of course we were able to use the definition, our postulates, and the rules of logic to prove that it is indeed true.

The exercises below will help you review the concepts of this chapter.

Label the following statements true or false.

1. If $AB \cong CD$, then $A = C$ and $B = D$.
2. If $A = C$ and $B = D$, then $AB \cong CD$.
3. If $AB \cong CD$ or $AB < CD$, then $AB \leqq CD$.
4. If A, B, and C are points on a line l and A', B', and C' are points on a line l', and $AB \cong A'B'$, and $BC \cong B'C'$, then $AC \cong A'C'$.
5. If AB is any segment and A' is any point on a line l', then there are exactly two points B' and B'' on l' such that $A'B'$ and $A'B''$ are congruent to AB.
6. If AB is not less than CD, then it must be greater than CD.
7. If RS is neither greater than nor less than TU, then $RS \cong TU$.
8. If A, B, C, and D are points in this order on a line l and $AC \cong BD$, then $AB \cong CD$.
9. If A, B, C, and D are points in this order on a line l and $AC > BD$, then $AB < CD$.
10. If A, B, C, and D are points in this order on a line l and $AC < BD$, then $AB \leqq CD$.

This chapter has included three postulates, two definitions, and four theorems. The following statements are all true. Designate a postulate or definition or theorem which could be used in justifying each statement.

11. For every line segment AB, it is true that $AB \cong AB$.
12. If $AB \cong CD$ and ABX, then $AX > CD$.
13. If AB is not less than CD, then $AB \geqq CD$.

14. If $AB \cong CD$ and $CD \cong EF$ and $XY > AB$, then $XY > EF$.

15. If $RS \cong TU$ and $XY \cong TU$, then $RS \cong XY$.

16. If ABC and BCD, then $AC < AD$.

17. If $AB \leqq CD$, then there is no point E such that CDE and $AB \cong CE$.

18. If in triangle ABC we have side AC greater than side AB, then there is a point E on side AC such that $AE \cong AB$.

19. If ABC and $AB \cong BC$ and XYZ and $XY \cong YZ \cong AB$, then $AC \cong XZ$.

20. If $A = X$ and $B = Y$, then $AY \cong BX$.

ALGEBRA REVIEW

In algebra the relations of "greater than" and "less than" are defined for pairs of numbers. We say that $5 > 3$ because $5 = 3 + 2$, and 2 is a positive number. More generally we make the following definitions:

(1) If a, b, and c are numbers and $a = b + c$ and c is positive, then we say that a is greater than b and write $a > b$.

(2) If a, b, and c are numbers and $a = b - c$ and c is a positive number, then we say that a is less than b and write $a < b$.

For example, since $-\frac{3}{2} = -\frac{5}{2} + \frac{2}{2}$, we say $-\frac{3}{2} > -\frac{5}{2}$; and since $0 = \frac{3}{4} - \frac{3}{4}$ we say that $0 < \frac{3}{4}$.

1. Use the first definition above to prove that $0 > -\frac{4}{3}$.

2. Use the second definition to prove that $-\frac{4}{3} < 0$.

3. Use the two definitions to prove that if $a > b$ then $b < a$.

4. If $a = b + 1$ and $b = c + 2$, prove that $a > c$.
5. Prove that if $a > b$ and $b > c$, then $a > c$.
6. If $a = b - 2$ and $b = c - 3$, then prove that $a < c$.
7. Prove that if $a < b$ and $b < c$, then $a < c$.
8. Show that if $a = b + 3$, then $3a > 3b$ and $-2a < -2b$.
9. Prove that if $a > b$, then $ca > cb$ if c is positive and $ca < cb$ if c is negative.
10. Prove that if $a < b$, then $ca < cb$ if c is positive and $ca > cb$ if c is negative.